

Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT305	Jimin XIAO	Intelligent Science	3209

Final Exam 2022/2023

BACHELOR DEGREE – Year 4

Machine Learning

TIME ALLOWED: 3 Hours 0 Minutes

INSTRUCTIONS TO CANDIDATES

- 1、 Total marks available are 100.**
- 2、 Answer all questions.**
- 3、 The number in the column on the right indicates the approximate marks for each section.**
- 4、 In general, it is particularly important to give reasons for your answer. Only partial marks will be awarded for correct answers with inadequate reasons.**
- 5、 Answer should be written in the answer booklet(s) provided.**
- 6、 The university approved calculator - Casio FS82ES/83ES can be used.**
- 7、 Only solutions written in English will be accepted.**

Answer All Questions

1. Pooling units take n values x_i , $i \in [1, n]$ and compute a scalar output whose value is invariant to permutations of the inputs.

(1) The Lp-pooling module takes positive inputs and computes $y = (\sum_i x_i^p)^{\frac{1}{p}}$, assuming we know that

$$y' = \frac{\partial L}{\partial y}, \text{ what is } x'_i = \frac{\partial L}{\partial x_i} ?$$

(2) The log-average module computes $y = \frac{1}{\beta} \ln(\frac{1}{n} \sum_i \exp(\beta x_i))$, assuming we know that $y' = \frac{\partial L}{\partial y}$, what

$$\text{is } x'_i = \frac{\partial L}{\partial x_i} ?$$

Total

16 (each 8)

2. Recall two linear classification methods we considered:

Model 1:

$$y = \mathbf{w}^T \mathbf{x} + b$$

$$\mathcal{L}_{SE}(y, t) = \frac{1}{2} (y - t)^2$$

Model 2:

$$z = \mathbf{w}^T \mathbf{x} + b$$

$$y = \sigma(z)$$

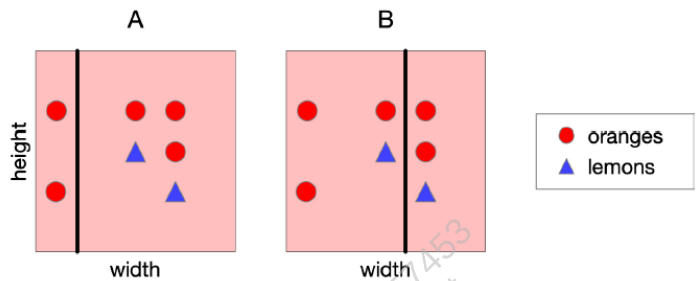
$$\mathcal{L}_{SE}(y, t) = \frac{1}{2} (y - t)^2$$

Here, σ denotes the logistic function (sigmoid function), and the target label t take values in $\{0, 1\}$. Briefly explain our reason for preferring Model 2 to Model 1.

Total

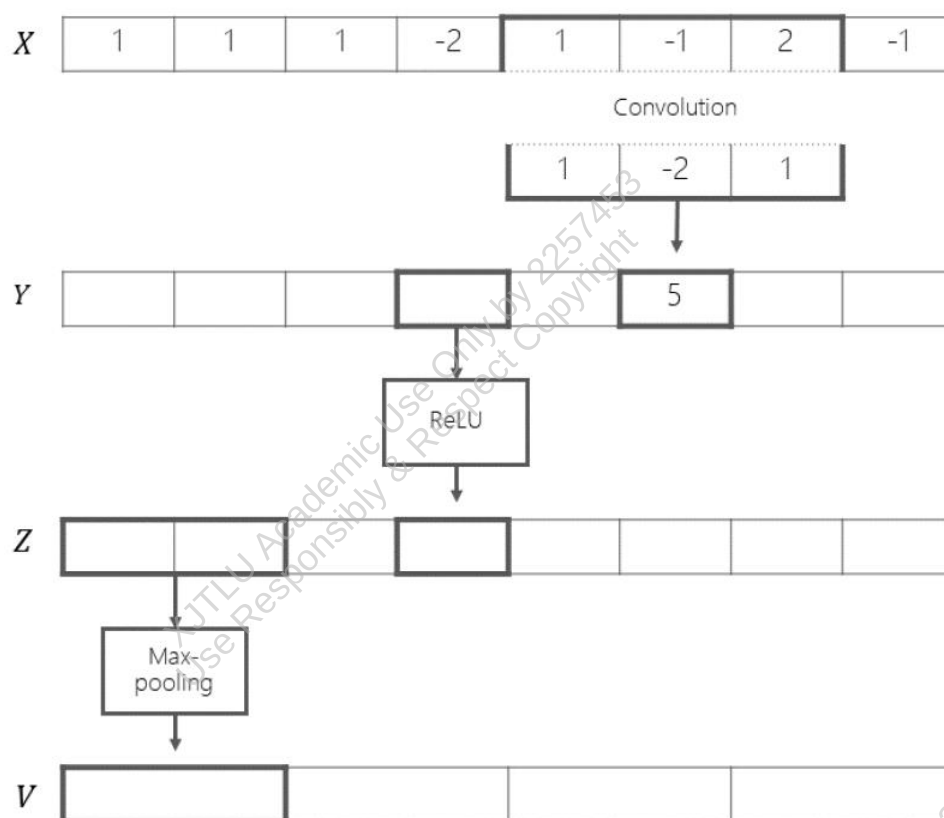
14

3. We are constructing a decision tree. Considering the following data, we would like to split them on width for one level. There are 2 options to split them as in the following figure. Please select the better one from the perspective of information gain.



Total
15

4. The sub-system shown in the following figure has a convolution unit (the convolution kernel is 1×3), followed by a ReLU, and then followed by max-pooling unit. In the following, we assume that the convolution operation uses zero padding.



Sub-system with convolution, ReLU, then followed by Max-Pooling unit.

- (1) Fill the empty elements (i.e., all the empty elements) in the vector Y .
- (2) Fill the empty elements (i.e., all the empty elements) in the vector Z .
- (3) Fill the empty elements (i.e., all the empty elements) in the vector V .
- (4) How many learnable parameters does this sub-system have?
- (5) Given the pictorial representation shown in the above figure, what is the stride size of the convolution unit?

Total

20 (each 4)

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5. You want to train the following model using gradient descent. Here, the input x and target t are both scalar-valued.

$$z = \omega_0 + \omega_1 x + \omega_2 x^2 + \omega_3 x^3$$

$$y = 1 + e^{\max(z, 0)}$$

$$L = \frac{1}{2}(\ln y - \ln t)^2$$

Determine the back propagation rules which will let you compute the loss derivative of $\bar{\omega}_2 = \frac{\partial L}{\partial \omega_2}$ (the variables with the bar are the gradient of the variables). Your equations should refer to previously computed values (e.g. your formula for $\bar{z}$ should be a function of $\bar{y}$). You do not need to show your derivation. The dummy step has been filled in for you.

$$\bar{\mathcal{L}} = 1$$

$$\bar{y} =$$

$$\bar{z} =$$

$$\bar{\omega}_2 =$$

Total
15 (each 5)

6. In this question, you will derive the E-M update rules for a univariate Gaussian mixture model (GMM) with two mixture components. Unlike the GMMs we covered in the course, the mean μ will be shared between the two mixture components, but each component will have its own standard deviation σ_k (σ_0 and σ_1). The mixture component is indicated by a latent variable $z \in \{0,1\}$. The model is defined as follows:

$$z \sim \text{Bernoulli}(\theta) \text{ (i.e., } P(z = 1|\theta) = \theta, P(z = 0|\theta) = 1 - \theta)$$

$$x | z = k \sim \mathcal{N}(\mu, \sigma_k) \text{ for } k \in \{0,1\}$$

The parameters of the model are θ , μ , σ_0 , and σ_1 . Suppose we observe a dataset $\{x^{(i)}\}_{i=1}^N$.

For reference, the PDF of the Gaussian distribution is as follows:

$$\mathcal{N}(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- (1) Write the **complete** data log-likelihood for this model. (In this case, we assume that $z^{(i)}$ is known for every $x^{(i)}$, thus our dataset is $\{z^{(i)}, x^{(i)}\}_{i=1}^N$)
- (2) In the E-step, for each data point $x^{(i)}$, we need to compute the posterior probability $r^{(i)} = \Pr(z^{(i)} = 1 | x^{(i)})$. Please Give the formula for $r^{(i)}$. In your formula, you may use $\mathcal{N}(x^{(i)}; \mu, \sigma)$ to denote the Gaussian PDF, rather than writing it out explicitly.

$$r^{(i)} =$$

- (3) Write out the expected complete data log-likelihood, i.e., the objective that is to be maximized in the M-step. It should be expressed in terms of the $r^{(i)}$ and the Gaussian PDF $\mathcal{N}(x^{(i)}; \mu, \sigma)$.
- (4) Derive the M-step update rule for μ by maximizing this objective with respect to μ . (In this step, the σ_k are fixed to their previous values.)

Total
20 (each 5)

The end of the paper